Lecture 24: Integer Optimization

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Announcements

- Homework 9 due on May 10th at noon
- Next lecture on May 6th

Recap: Integer Optimization

Integer linear program: A linear program with some decision variables having to be integers.

- Binary integer programs
- Mixed integer programs

We showed some examples of IPs

- Knapsack problem
- Facility location problem
- Matching problem

Using integer variables can allow us to represent more complicated feasible regions

Recap: Hardness of IP

In general, IP is very hard to solve

- No polynomial-time algorithm is known to solve general IP

LP relaxation

- Relax (ignore) all integral constraints and solve it as a linear program
- Relaxation of binary constraint $x \in \{0, 1\}$: $0 \leq x \leq 1$
- Provide bounds on the optimal value of IP (upper bound for maximization and lower bound for minimization)
- It can also be used to provide performance bound for any *feasible* solution to the IP
- Integrality gap

$$|v^{LP} - v^{IP}|$$

Some Good Cases

There are some cases in which solving the LP relaxation can give one very good solution

Theorem

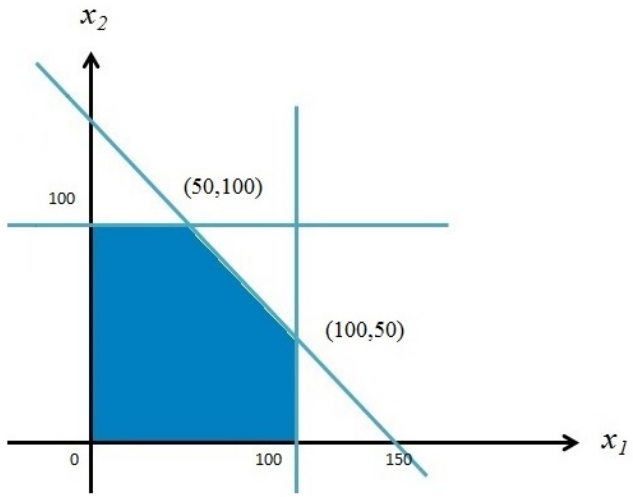
If the optimal solution to the LP relaxation is integral, then the solution must be optimal to the IP problem.

When Does LP Have Integral Solutions

Remember for LP, there must exist an optimal solution that is a basic feasible solution.

- If we can guarantee that every BFS is integral, then the LP must have an integer optimal solution

For example, in the production planning problem:



All the extreme points are integral. Therefore, there must exist an optimal solution that is integral (and simplex method will find it).

- This criterion has nothing to do with the objective function

Totally Unimodularity

We introduce a condition under which all BFS must be integral — totally unimodularity.

Definition

A matrix A is said to be *totally unimodular* if the determinant of each submatrix of A is either 0, 1, or -1.

Theorem

If the constraint matrix A is totally unimodular and $\mathbf{b}$ is integral, then all the BFS are integers and the LP must have an optimal solution that is an integral solution.

Totally unimodularity (TU)

- In order for A to be TU, it can only have 0, 1 or -1 entries
- But it needs much more

One Sufficient Condition for TU

Theorem

Let A be an $m \times n$ matrix. Then the following conditions together are sufficient for A to be totally unimodular:

1. *Every column of A contains at most two non-zero entries;*
2. *Every entry in A is 0, $+1$, or -1;*
3. *The rows of A can be partitioned into two disjoint sets B and C such that*
   (a) *If two non-zero entries in a column of A have the same sign, then the row of one entry is in B, and the row of the other in C;*
   (b) *If two non-zero entries in a column of A have opposite signs, then the rows are both in B, or both in C.*

Example

Recall the matching problem:

$$\begin{aligned} \text{maximize} \quad & \sum_{i=1}^{n} \sum_{j=1}^{n} v_{ij} x_{ij} & \\ \text{s.t.} \quad & \sum_{i=1}^{n} x_{ij} = 1 & \forall j \\ & \sum_{j=1}^{n} x_{ij} = 1 & \forall i \\ & x_{ij} \in \{0, 1\} & \end{aligned}$$

The constraint matrix is TU.
An example when $n = 2$ (in the order of $x_{11}, x_{12}, x_{21}, x_{22}$):

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

One can choose $B = \{1, 2\}$, $C = \{3, 4\}$. The condition holds and A is TU.

Matching Problem

Therefore, to solve the matching problem, we can safely relax the binary constraints to $0 \leq x_{ij} \leq 1$. It is guaranteed that there exists an integral optimal solution to the LP (and the simplex method can find it).

- This result also applies to transportation problems, assignment problems which have the same constraint matrix as the matching problem
- There are some network flow problems which also have the TU properties

Another Example

Consider the following matrix A:

$$A = \begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \end{bmatrix}$$

Then we can make the partition to be $B = \{1, 2, 3, 4\}$, $C = \phi$.
The condition holds and A is TU

Solve IP

TU property is uncommon in practice - if one solves the LP relaxation and find the optimal BFS is not integral, it means that TU doesn't hold

- We need to have methods to solve IP systematically (and efficiently)
- The most commonly applied technique, as we will discuss in the following, is the *branch-and-bound* method

The Idea

Consider the following example

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0, x_1, x_2 \in \mathbb{Z} \end{aligned}$$

- We solve the LP relaxation and get $\mathbf{x}^* = (15/4, 9/4)$
- x_1 is not feasible (we can't allow $15/4$ in the solution)

Solution: Consider two subproblems:

- One with an additional constraint $x_1 \leq 3$.
- One with an additional constraint $x_1 \geq 4$.

The optimal solution to the IP must still be in one of the subproblems, but solutions with $3 < x_1 < 4$ are eliminated

Generally

First, we solve the LP relaxation of the IP:

- If the solution is already integral, then it is optimal to the IP

If the optimal solution to the LP relaxation is $\mathbf{x}^*$ and x_i^* is fractional, then branch the problem into the following two:

1. One with an added constraint that $x_i \leq \lfloor x_i^* \rfloor$, we call this (S1)
2. One with an added constraint that $x_i \geq \lceil x_i^* \rceil$, we call this (S2)

Here $\lfloor \cdot \rfloor$ means rounding down, and $\lceil \cdot \rceil$ means rounding up.

- $\lfloor 3.75 \rfloor = 3$, $\lceil 3.75 \rceil = 4$

What Happens after Branching

We get two IP's after branching.

- We then solve (S1) and (S2) and assume we can get the optimal solutions $\mathbf{x}_1^*$ and $\mathbf{x}_2^*$ with optimal values v_1^* and v_2^* (both (S1) and (S2) are still integer programs).

Claim

- *If $v_2^* \leq v_1^*$, then $\mathbf{x}_1^*$ is the optimal solution to the original IP*
- *If $v_1^* \leq v_2^*$, then $\mathbf{x}_2^*$ is the optimal solution to the original IP*

The claim is true because the union of the feasible regions in each branch equals the feasible region of the original problem

How to solve (S1) and (S2)?

- Use the same idea (solve LP relaxation and further branch)

Bounding

For each branch, we can construct an upper bound and a lower bound for the problem (assume we are solving a maximization problem):

- Upper bound: The LP relaxation solution will be an upper bound — The objective value of any integral solution from this node must be lower than the optimal value of the LP relaxation
- Lower bound: The objective value of any feasible (integral) solution is a lower bound for the optimal value — The optimal solution of the IP must be no less than the objective value achieved by any feasible solution
- When at a certain node, the optimal value of the LP relaxation of this branch is even less than the current lower bound. Then we should abandon this branch
- These results will be the opposite if we are minimizing

We will use the bounding steps to *prune* unnecessary computations

Example

Consider the earlier example

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0, x_1, x_2 \in \mathbb{Z} \end{aligned}$$

- We solve the LP relaxation and get $\mathbf{x}^* = (15/4, 9/4)$
- We branch for x_1

We solve two subproblems:

- One with an additional constraint $x_1 \leq 3$. We call it (S1)
- One with an additional constraint $x_1 \geq 4$. We call it (S2)

Example Continued

Now we solve (S1) and (S2) respectively

- We solve the LP relaxation of (S1), and get the optimal solution is $(3, 3)$. This is an integral solution, so we are done with this branch (the optimal value is 39)
- We solve the LP relaxation of (S2), the optimal solution is $(4, 1.8)$ (the optimal value is 41).

The solution to (S2) is not integral. We have to do further branching:

- We add a constraint that $x_2 \leq 1$ (S3)
- We add a constraint that $x_2 \geq 2$ (S4)

Example Continued

S3:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1 \geq 4, x_2 \leq 1 \quad x_1, x_2 \geq 0, x_1, x_2 \in \mathbb{Z} \end{aligned}$$

S4:

$$\begin{aligned} \text{maximize} \quad & 8x_1 + 5x_2 \\ \text{subject to} \quad & 9x_1 + 5x_2 \leq 45 \\ & x_1 + x_2 \leq 6 \\ & x_1 \geq 4, x_2 \geq 2 \quad x_1, x_2 \geq 0, x_1, x_2 \in \mathbb{Z} \end{aligned}$$

One can easily see that (S4) is not feasible. Thus we don't need to further consider that.

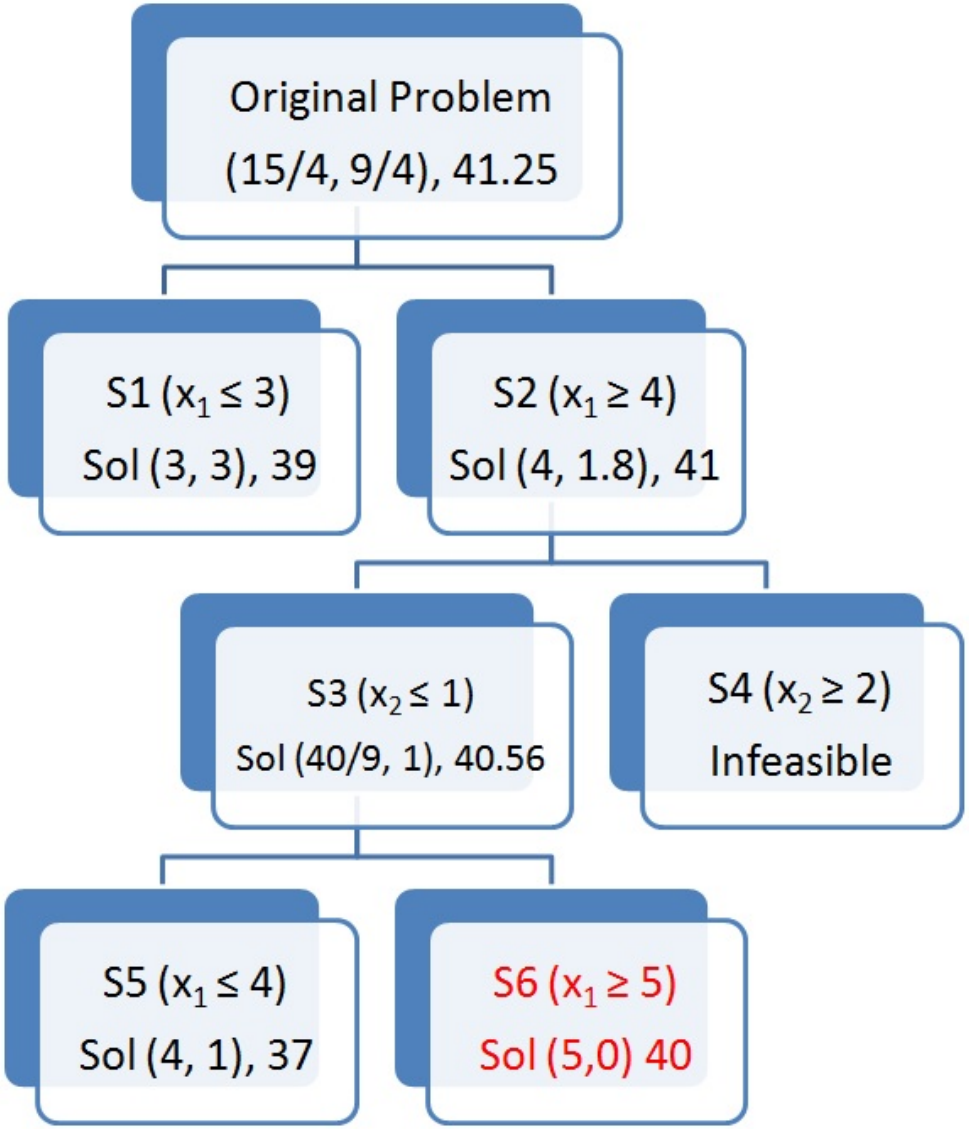
Solve (S3)

We solve (S3) and the optimal solution is $(40/9, 1)$, we have to do further branching:

- Add a constraint $x_1 \leq 4$ (S5)
- Add a constraint $x_1 \geq 5$ (S6)

For (S5), x_1 has to be 4 and the optimal solution is $(4, 1)$ with objective value 37 (already integer, so don't need to further branching).

For (S6), the optimal solution is $(5, 0)$, the objective value is 40.


Original Problem
(15/4, 9/4), 41.25
S1 ($x_1 \leq 3$)
Sol (3, 3), 39
S2 ($x_1 \geq 4$)
Sol (4, 1.8), 41
S3 ($x_2 \leq 1$)
Sol (40/9, 1), 40.56
S4 ($x_2 \geq 2$)
Infeasible
S5 ($x_1 \leq 4$)
Sol (4, 1), 37
S6 ($x_1 \geq 5$)
Sol (5,0) 40